

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: VIth

Subject Code & Name: BTCVOE605H Applications of Remote Sensing & GIS

Max Marks: 60

Date: 21/07/2023

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Draw a suitable diagram wherever necessary.

Marks

Q. 1 Solve Any Two of the following.

- | | |
|---|----------|
| A) Write short notes on: - | 6 |
| a. Rayleigh scattering | |
| b. Atmospheric window | |
| c. Use of low frequency waves | |
| B) Write a detailed note on the procedure of the remote sensing technique. | 6 |
| C) Explain the electromagnetic spectrum and its role in remote sensing. | 6 |

Q.2 Solve Any Two of the following.

- | | |
|---|----------|
| A) Write a short note on: - | 6 |
| a. Swath | |
| b. Whisk-broom scanning system | |
| c. NAVIC system | |
| B) Write a difference between active and passive remote sensing. | 6 |
| C) Enumerate a microwave radar sensing system with its limitations and advantages. | 6 |

Q. 3 Solve Any Two of the following.

- | | |
|--|----------|
| A) Write a short note on: - | 6 |
| a. N. D. V.I. | |
| b. Analogue image | |
| c. Radiometric correction | |
| B) Write a detailed note on spectral signature characteristic features of vegetation and influencing factors. | 6 |
| C) Explain in detail the supervised classification of imageries with the procedure and its disadvantages. | 6 |

Q.4 Solve Any Two of the following.

- A)** Write a short note on: - **6**
- a. Triangular Irregular Network (TIN)
 - b. D.E.M.
 - c. Attribute data
- B)** Enumerate the difference between raster data and vector data. **6**
- C)** Write a detailed note on Geographic Information System with its advantages and limitations. **6**

Q. 5 Solve Any Two of the following.

- A)** Write a short note on: - **6**
- a. Universal Transverse Mercator
 - b. Geoid
 - c. Longitude
- B)** Write a detailed note on Geographic Coordination System with its strengths and weaknesses. **6**
- C)** Explain in detail about various techniques used for approximation of Earth. **6**

***** End *****